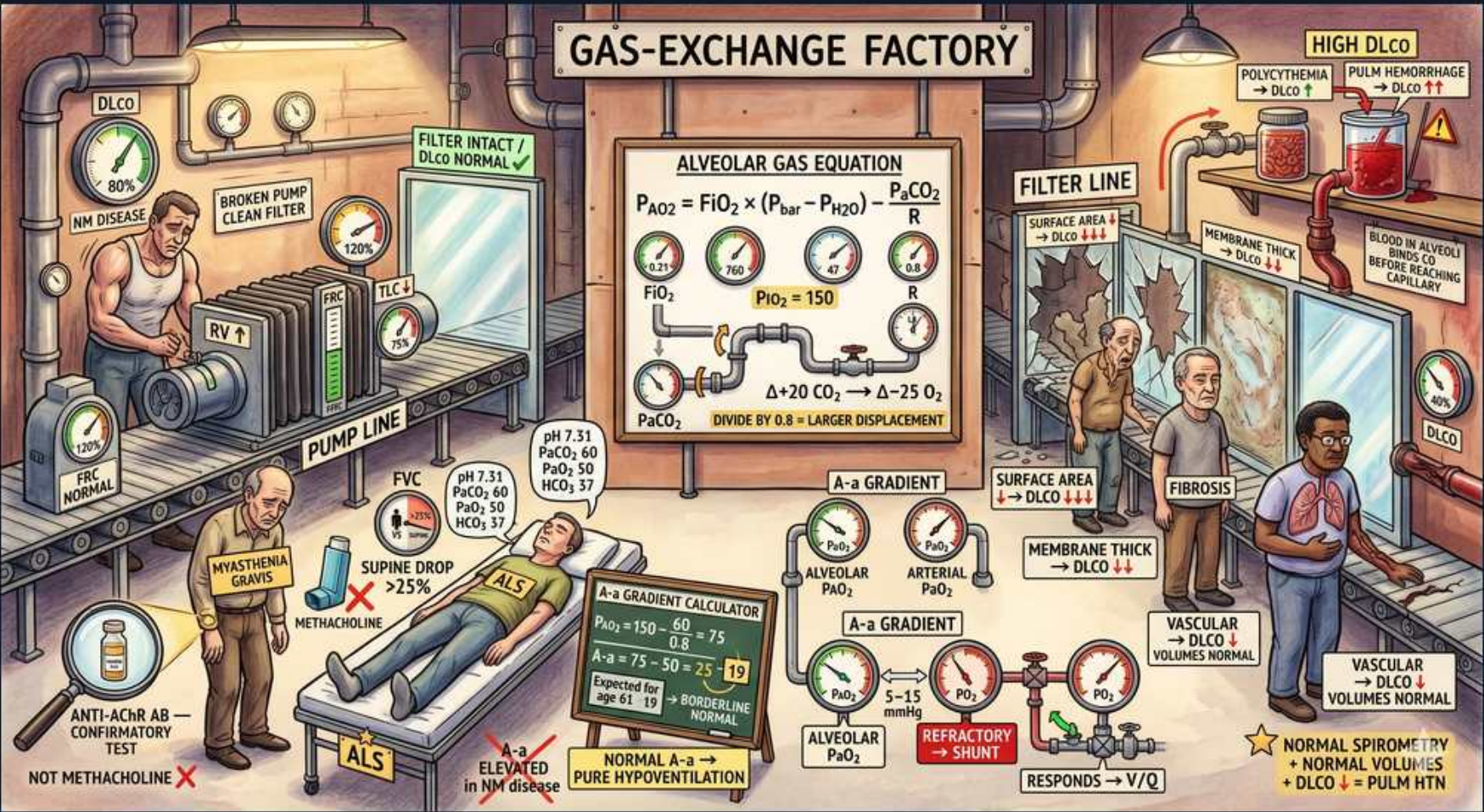


Respiratory Disturbance — Hypoxemia, A-a Gradient & Shunt (Gas-Exchange Factory)



1. Alveolar gas equation

WHAT YOU SEE

Big formula board: $PAO_2 = FiO_2 (P_{bar} - PH_2O) - PaCO_2/R$

WHAT IT ENCODES

$PAO_2 = FiO_2 \times (P_{atm} - PH_2O) - PaCO_2/R$. On room air at sea level this simplifies to $PAO_2 \approx 150 - PaCO_2/0.8$ (since $0.21 \times (760 - 47) \approx 150$ and $R \approx 0.8$). PAO_2 is the calculated ALVEOLAR oxygen you subtract the measured arterial PaO_2 from to get the A–a gradient.

BOARD TRAP

Forgetting to divide $PaCO_2$ by R (≈ 0.8), or forgetting to subtract water vapor (47 mmHg), undercalculates expected PAO_2 and misclassifies the A–a gradient.

2. $PIO_2 = 150$

WHAT YOU SEE

Gauge reading 150 from $FiO_2 \times (P_{bar} - PH_2O)$

WHAT IT ENCODES

Inspired/alveolar $PO_2 \approx 150$ mmHg on room air at sea level. At ALTITUDE the barometric pressure falls, so inspired PO_2 falls → hypoxemia with a NORMAL A–a gradient (the lung is fine; the air is thin). Supplemental O_2 corrects it readily.

BOARD TRAP

High-altitude (and any low- FiO_2) hypoxemia is NOT a shunt or diffusion problem — the A–a gradient stays NORMAL because both alveolar and arterial O_2 fall together.

3. NM disease / clean filter

WHAT YOU SEE

Strongman with 80% DLCO meter, 'broken pump clean filter'

WHAT IT ENCODES

Neuromuscular weakness causes HYPOVENTILATION — the muscular 'pump' fails but the gas-exchange 'filter' (alveolar membrane) is intact, so the A–a gradient is NORMAL. $PaCO_2$ rises and, because $PAO_2 = 150 - PaCO_2/0.8$, the rising CO_2 mathematically lowers PAO_2 and thus PaO_2 .

BOARD TRAP

Pure hypoventilation has a NORMAL A–a gradient — do not call it a shunt or fibrosis. The two normal-gradient causes of hypoxemia are hypoventilation and low FiO_2 /altitude.

4. Myasthenia gravis

WHAT YOU SEE

Vial labeled Myasthenia Gravis with anti-AChR / methacholine note

WHAT IT ENCODES

Myasthenia gravis causes a restrictive/hypoventilatory pattern from respiratory muscle weakness; diagnose with anti-acetylcholine-receptor (anti-AChR) antibodies (and anti-MuSK if seronegative), plus repetitive nerve stimulation/single-fiber EMG.

BOARD TRAP

Methacholine challenge tests for ASTHMA (airway hyperresponsiveness), NOT for neuromuscular weakness — don't pick it to work up MG.

5. Supine FVC drop >25%

WHAT YOU SEE

FVC gauge with 'supine drop >25%' tag

WHAT IT ENCODES

A fall in FVC of >25% (often cited >10–25%) from upright to supine indicates BILATERAL diaphragm weakness (the abdominal contents push the weak diaphragm up when lying flat). Seen in ALS, bilateral phrenic palsy, and advanced neuromuscular disease; presents as orthopnea.

BOARD TRAP

A large supine FVC drop points to diaphragmatic/neuromuscular disease, not parenchymal lung disease — it's the classic clue to bilateral diaphragm paralysis.

6. ALS patient

WHAT YOU SEE

Patient lying on bench labeled ALS

WHAT IT ENCODES

ALS leads to progressive respiratory muscle failure → pure hypoventilation: rising $PaCO_2$ with a NORMAL A–a gradient. Respiratory failure (often nocturnal hypoventilation first) is the leading cause of death; treat with noninvasive ventilation (BiPAP).

BOARD TRAP

Hypercapnia + a NORMAL A–a gradient = a VENTILATION problem, not an oxygenation problem — the fix is supporting ventilation (NIV), and supplemental O_2 alone can worsen hypercapnia.

7. A–a gradient calculator

WHAT YOU SEE

Box: $150 - 50 - 75 = 19$ with expected range

WHAT IT ENCODES

A–a gradient = $PAO_2 - PaO_2$. Normal estimate $\approx (age/4) + 4$, and it widens physiologically with age. It is the central tool that separates the NORMAL-gradient causes (hypoventilation, low FiO_2 /altitude) from the WIDENED-gradient causes (V/Q mismatch, shunt, diffusion limitation).

BOARD TRAP

Using a single fixed cutoff (e.g., '10') for every patient is wrong — the normal A–a gradient rises with age, so a value of 19 may be normal in an older adult.

8. A–a gradient gauges

WHAT YOU SEE

Two dials: alveolar vs arterial O_2

WHAT IT ENCODES

Interpretation: a WIDENED A–a gradient = V/Q mismatch, right-to-left shunt, or diffusion defect (a problem in the lung between alveolus and capillary). A NORMAL A–a gradient = hypoventilation or low FiO_2 /altitude (the lung itself is fine).

BOARD TRAP

A widened gradient EXCLUDES pure hypoventilation/low- FiO_2 as the sole cause — there must be intrinsic lung pathology contributing.

9. Normal A–a / pure hypoventilation

WHAT YOU SEE

Banner: 'Normal A–a = pure hypoventilation', elevated in NM disease

WHAT IT ENCODES

Hypoxemia + elevated PaCO₂ + NORMAL A–a gradient = pure hypoventilation: CNS depression (opioids, sedatives, central hypoventilation), neuromuscular disease, or chest-wall disorders. Correcting ventilation corrects the hypoxemia.

BOARD TRAP

Don't attribute pure-hypoventilation hypoxemia to lung parenchyma — if the gradient is normal, the alveolar-capillary unit is working and the problem is ventilatory drive/mechanics.

10. Refractory shunt

WHAT YOU SEE

Red 'REFRACTORY SHUNT' tag on a valve

WHAT IT ENCODES

A right-to-left SHUNT (blood bypasses ventilated alveoli) is the V/Q extreme of zero ventilation. Causes: ARDS, lobar pneumonia, atelectasis, pulmonary edema, AVMs, and intracardiac R→L shunts. It is REFRACTORY to supplemental O₂ because shunted blood never contacts alveolar gas.

BOARD TRAP

Failure to correct hypoxemia with 100% O₂ = SHUNT, NOT V/Q mismatch. This 100%-O₂ test is the classic way to distinguish the two on the boards.

11. Responds to V:Q / O₂

WHAT YOU SEE

Gauge 'responds to V:Q' near O₂ supply

WHAT IT ENCODES

V/Q mismatch is the MOST COMMON cause of hypoxemia (COPD, asthma, PE, pneumonia, atelectasis). Because some ventilation reaches the affected units, raising FiO₂ boosts the alveolar O₂ in low-V/Q regions and CORRECTS the hypoxemia.

BOARD TRAP

If hypoxemia improves with supplemental O₂, it is V/Q mismatch — true shunt would NOT correct. (V/Q mismatch is the default/most-common answer for unexplained hypoxemia.)

12. Surface area ↑→ DLCO ↑

WHAT YOU SEE

Filter line: larger surface area → DLCO up

WHAT IT ENCODES

DLCO is governed by alveolar-capillary SURFACE AREA, MEMBRANE thickness, and HEMOGLOBIN/pulmonary blood volume. More functioning surface area or more Hgb → higher DLCO; destroyed/thickened membrane or less Hgb → lower DLCO.

BOARD TRAP

Anemia falsely LOWERS DLCO; polycythemia and alveolar hemorrhage falsely RAISE it — always interpret DLCO after correcting for hemoglobin.

13. Fibrosis (membrane thick ↓DLCO)

WHAT YOU SEE

Patient labeled fibrosis, membrane thickening lowers DLCO

WHAT IT ENCODES

ILD/fibrosis thickens and destroys the alveolar-capillary membrane → DIFFUSION limitation → low DLCO and a WIDENED A–a gradient. Diffusion-limited hypoxemia classically appears or worsens with exercise (less transit time for equilibration) and does respond to supplemental O₂.

BOARD TRAP

Low DLCO + restriction = ILD/fibrosis, NOT obesity or neuromuscular disease (those keep a normal DLCO).

14. Vascular ↓DLCO / Pulm HTN

WHAT YOU SEE

Sign: vascular → DLCO down, volumes normal

WHAT IT ENCODES

Isolated LOW DLCO with NORMAL spirometry and NORMAL lung volumes points to pulmonary VASCULAR disease — pulmonary arterial hypertension or chronic thromboembolic disease (loss of the capillary bed for CO uptake).

BOARD TRAP

Normal volumes + normal FEV₁/FVC but an ISOLATED low DLCO = pulmonary vascular disease (PAH/CTEPH), not parenchymal or airway disease — a very high-yield pattern.

15. High DLCO causes

WHAT YOU SEE

Top-right banner: polycythemia, pulmonary hemorrhage ↑DLCO

WHAT IT ENCODES

High DLCO causes: diffuse alveolar hemorrhage, polycythemia, left-to-right shunt / early CHF (increased pulmonary blood volume), asthma, exercise, and the supine position.

BOARD TRAP

A HIGH DLCO is a clue to alveolar hemorrhage or polycythemia — not a normal incidental finding to ignore; pair it with hemoptysis/anemia (hemorrhage) or elevated Hct (polycythemia).